

Dividing Complex Numbers

Answer Key

Algebra 2

Quick Answers

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|--------------|-----------------------|----------------|
| 1. $3 - 4i$ | 5. $6 - 2i$ amps | 9. $2 + 5i$ |
| 2. $6 - 8i$ | 6. $-4 - 2i$ | 10. (a) = 2, — |
| 3. $-4 - 3i$ | 7. (a) = $3 + 4i$, — | |
| 4. $5 - 2i$ | 8. $7 - 4i$ ohms | |
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What is verified

10 of 12 answers machine-checked against SymPy, across 10 problems.

— marks an answer only you can judge — problems 7, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 2

HSN-CN.A.3 – problems 1, 2, 3, 4, 5, 6, 8, 9, 10

HSN-CN.B.4 – problem 7

Difficulty 1–5 of 5 across 12 tagged checks.

1. Simplify $\frac{8 + 6i}{2i}$.

Solution: The denominator is purely imaginary, so its conjugate is $-2i$. Multiplying top and bottom by $-2i$ clears the i from the bottom.

$$\frac{8 + 6i}{2i} \cdot \frac{-2i}{-2i} = \frac{-16i - 12i^2}{-4i^2} = \frac{12 - 16i}{4}$$

Dividing both parts by 4 gives the standard form.

$3 - 4i$

2. Simplify $\frac{50}{3 + 4i}$.

Solution: The conjugate of $3 + 4i$ is $3 - 4i$, and $(3 + 4i)(3 - 4i) = 9 + 16 = 25$.

$$\frac{50}{3 + 4i} \cdot \frac{3 - 4i}{3 - 4i} = \frac{150 - 200i}{25} = 6 - 8i$$

$6 - 8i$

3. Simplify $\frac{9 - 12i}{3i}$.

Solution: The conjugate of $3i$ is $-3i$; the denominator becomes $(3i)(-3i) = -9i^2 = 9$.

$$\frac{9 - 12i}{3i} \cdot \frac{-3i}{-3i} = \frac{-27i + 36i^2}{9} = \frac{-36 - 27i}{9}$$

$-4 - 3i$

4. Simplify $\frac{16 + 11i}{2 + 3i}$.

Solution: Multiply by the conjugate $2 - 3i$. Denominator: $(2 + 3i)(2 - 3i) = 4 + 9 = 13$.

$$\begin{aligned}(16 + 11i)(2 - 3i) &= 32 - 48i + 22i - 33i^2 = 65 - 26i \\ \frac{65 - 26i}{13} &= 5 - 2i\end{aligned}$$

$5 - 2i$

5. An AC source supplies 20 volts across a component of impedance $3 + i$ ohms. Current equals voltage divided by impedance. Find the current in amps.

Solution: The conjugate of $3 + i$ is $3 - i$, and $(3 + i)(3 - i) = 9 + 1 = 10$.

$$\frac{20}{3 + i} \cdot \frac{3 - i}{3 - i} = \frac{60 - 20i}{10} = 6 - 2i$$

The real part is the in-phase current and the imaginary part records the phase shift, which is why the answer is reported as a complex number rather than a single magnitude.

$6 - 2i$ amps

6. Simplify $\frac{-8 + 6i}{1 - 2i}$.

Solution: Multiply by the conjugate $1 + 2i$; the denominator becomes $1 + 4 = 5$.

$$\begin{aligned}(-8 + 6i)(1 + 2i) &= -8 - 16i + 6i + 12i^2 = -20 - 10i \\ \frac{-20 - 10i}{5} &= -4 - 2i\end{aligned}$$

$-4 - 2i$

7. A drawing program multiplies each point by $3 + 4i$. One transformed point landed at $-7 + 24i$. (a) Which point did it start from? (b) Plot that point and label the axes.

Solution (a): Undo a multiplication by dividing. The conjugate of $3 + 4i$ is $3 - 4i$, and $(3 + 4i)(3 - 4i) = 9 + 16 = 25$.

$$\begin{aligned} (-7 + 24i)(3 - 4i) &= -21 + 28i + 72i - 96i^2 = 75 + 100i \\ \frac{75 + 100i}{25} &= 3 + 4i \end{aligned}$$

Check by multiplying back: $(3 + 4i)(3 + 4i) = 9 + 24i - 16 = -7 + 24i$.

$$\boxed{3 + 4i}$$

Part (b) — instructor-judged. The plotted point should sit 3 units along the real (horizontal) axis and 4 units up the imaginary (vertical) axis, with both axes labelled. Grade the plot against whatever value the student wrote in part (a), so a correct plot of a wrong quotient still earns the plotting mark.

8. Voltage $26 + 13i$ volts, current $2 + 3i$ amps. Impedance equals voltage divided by current. Find the impedance in ohms.

Solution: Multiply by the conjugate $2 - 3i$; the denominator becomes $4 + 9 = 13$.

$$\begin{aligned} (26 + 13i)(2 - 3i) &= 52 - 78i + 26i - 39i^2 = 91 - 52i \\ \frac{91 - 52i}{13} &= 7 - 4i \end{aligned}$$

A negative imaginary part means the reactance is capacitive.

$$\boxed{7 - 4i \text{ ohms}}$$

9. Simplify $\frac{29i}{5 + 2i}$.

Solution: A purely imaginary numerator changes nothing about the method: multiply by the conjugate $5 - 2i$, whose product with the denominator is $25 + 4 = 29$.

$$\begin{aligned} 29i(5 - 2i) &= 145i - 58i^2 = 58 + 145i \\ \frac{58 + 145i}{29} &= 2 + 5i \end{aligned}$$

$$\boxed{2 + 5i}$$

10. Simplify $\frac{3 + i}{1 - i} + \frac{3 - i}{1 + i}$, then justify why the result had to be real.

Solution (a): Handle each quotient with its own conjugate. Both denominators become $1 + 1 = 2$.

$$\begin{aligned} \frac{3 + i}{1 - i} &= \frac{(3 + i)(1 + i)}{2} = \frac{2 + 4i}{2} = 1 + 2i \\ \frac{3 - i}{1 + i} &= \frac{(3 - i)(1 - i)}{2} = \frac{2 - 4i}{2} = 1 - 2i \end{aligned}$$

Adding the two results, the imaginary parts cancel.

$$\boxed{2}$$

Part (b) — instructor-judged. A full-credit justification says the two quotients are conjugates of one another — the second fraction is the first with the sign of every imaginary part reversed — and that adding a number to its own conjugate cancels the imaginary parts and leaves twice the real part. Showing $1 + 2i$ and $1 - 2i$ and stating that their imaginary parts are opposites earns full credit. Restating the answer 2 with no reason earns none.